

CoRoT-1b

R-2562

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-1_190129 · created 2026-08-26T23:16:24+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458512.8218 +/- 0.0041 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.085 +/- 0.04
Transit depth (Rp/Rs)^2	0.73 +/- 0.68 [%]
Orbital Inclination (inc)	88.59 +/- 2.83
Airmass coefficient 1 (a1)	1.17 +/- 0.27
Airmass coefficient 2 (a2)	-0.064 +/- 0.047
Scatter in the residuals of the lightcurve fit is	23.96 %
Best Comparison Star	#2 - [292, 419]
Optimal Aperture	1.25
Optimal Annulus	8.46
Transit Duration (day)	0.1056 +/- 0.0066

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.085 $\pm$ 0.04 vs 0.1388 (deviation 1.33 σ)
Duration fit vs published	0.1056 d vs 0.1046 d (deviation 0.15 σ)
Mid-time O-C	-8.2 min
Depth SNR	2.1
Residual scatter	23.96 %

Light curve

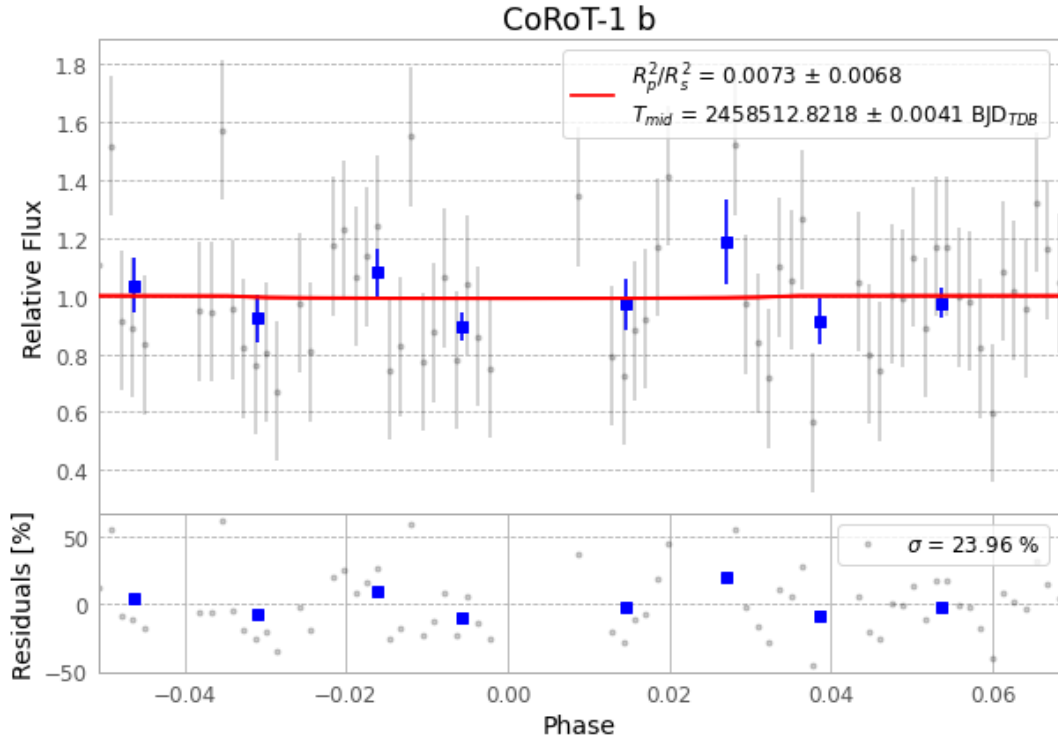


Figure 1 — FinalLightCurve_CoRoT-1 b_2019-01-28.png (SHA-256 3b008ef4d94e60df...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [364, 210], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 33c71513534c6ef1...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

87 FITS frames (57 MB), CoRoT-1190129054512.FITS → CoRoT-1190129100312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-1-190129.log (d65deeadb942...), FinalLightCurve_CoRoT-1 b_2019-01-28.png (3b008ef4d94e...), FinalLightCurve_CoRoT-1 b_2019-01-28.csv (88ba6b59ebca...)

Compute resources

Wall time 174.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)